

SANJAY AREPALLY

AWS & DATABRICKS CERTIFIED ENGINEER

Tampa, FL | +1 (813) 304 9504 | sanjay.arepally.data@gmail.com |

SUMMARY

Data Engineer with 4+ years of experience designing and owning scalable, production-grade data platforms end-to-end from ingestion and transformation to validation, monitoring, and performance optimization, while evolving solutions as business requirements change. Proven track record of translating complex business domains into well-structured data models, enforcing data quality and SLAs, and building reliable pipelines that power operational and executive decision-making. Experienced in Python, SQL, Spark, Databricks, AWS, dbt, Airflow, and modern cloud data ecosystems, with a strong focus on automation, system clarity, and long-term maintainability, and known for taking full ownership of data systems, collaborating closely with analytics, product, and engineering teams, and delivering data solutions that are fast, reliable, and trusted.

TECH STACK

- **Data Engineering & Modeling:** Data Modeling (Facts & Dimensions), Analytical Data Models, Metrics Definition, Data Acquisition & Ingestion, Batch Processing, Near Real-Time & Streaming Concepts, Schema Management
- **Programming & Processing:** Python, SQL, Spark (PySpark), Databricks
- **Orchestration & Transformation:** Airflow, dbt, Workflow Orchestration, Incremental Processing, Backfills
- **Cloud & Data Platforms:** AWS (S3, Glue, Redshift, Athena, EMR), Databricks, Apache Spark Ecosystem, Relational Databases (PostgreSQL)
- **Data Quality, Governance & Observability:** Data Validation Rules, Anomaly Detection, Data Quality Frameworks, Logging, Monitoring, Alerting, and Dataset Enablement
- **Software Engineering Practices:** CI/CD for Data Pipelines, Version Control (Git), Testing, Agile Development, Performance Optimization
- **Collaboration & Enablement:** Partnering with Product, Analytics & Data Science Teams, Requirements Clarification, Documentation, and Data Consumer Enablement.

CERTIFICATIONS

AWS Data Engineer – Associate | Databricks Data Engineer Associate | AWS Solutions Architect – Associate | Microsoft Power Platform PL-900 | Executive PG in Data Science & Analytics (IIT Roorkee) | Generative AI Certifications.

EXPERIENCE

Southwest Airlines, TX, USA | Data Engineer

June 2025 – Present

- Designed and maintained fact and dimension data models to track staffing and training completion, creating a single, reliable source of data used consistently by analytics, finance, and operations teams.
- Built and ran production ETL pipelines using Python and Spark to ingest, validate, and transform large volumes of structured data, supporting accurate reporting and downstream analytics.
- Added automated data quality checks, including schema validation, completeness checks, reconciliation logic, and anomaly detection, to catch data issues early and reduce recurring production problems by ~35%.
- Implemented incremental processing and safe backfill workflows so data stayed fresh while still allowing historical reprocessing without breaking downstream reports.
- Improved pipeline performance and reduced cloud costs by optimizing partitions, tuning queries, and adjusting compute usage across cloud data platforms and warehouses.
- Delivered certified, self-service datasets that allowed business and analytics teams to answer questions without engineering help, reducing ad-hoc requests and improving trust in reported metrics.
- Defined clear data SLAs for freshness, completeness, and accuracy, backed by monitoring, logging, and alerting to ensure predictable pipeline behavior and faster issue resolution.
- Set up CI/CD pipelines and deployment automation for data workflows, enabling safer releases, faster iteration, and more reliable production systems.
- Worked closely with analytics, data science, and product teams to turn changing business needs into scalable data models and pipelines, while ensuring existing consumers were not disrupted.

- Built and ran scalable data ingestion and transformation pipelines using Databricks, Spark, and orchestration tools to bring candidate data together from multiple ATS systems, reducing data refresh times from daily batches to near-real-time (4-hour) updates for recruiting and operations dashboards.
- Designed and maintained dimensional data models for recruiting data (candidates, offers, hires, and stage transitions), creating a single, consistent analytics layer used by HR, operations, and leadership teams.
- Implemented data governance and privacy controls using PySpark, including validation, masking, and schema enforcement, to protect PII and meet internal privacy standards while removing the need for manual data cleanup.
- Delivered certified, self-service datasets and access layers that allowed hiring managers and analysts to explore metrics on their own, significantly reducing ad-hoc SQL requests and ticket-based engineering work.
- Defined clear dataset SLAs for freshness, completeness, and accuracy, supported by automated monitoring and alerts to ensure reliable, morning-ready reports for executive and operational reviews.
- Set up an end-to-end pipeline monitoring system with logging, metrics, and anomaly detection to catch ingestion issues, schema changes, and data volume drops before they affect downstream dashboards.
- Modernized deployment workflows by adding CI/CD automation, version control, and automated testing, enabling safe, predictable pipeline updates with no downtime during business hours.

Flipkart, Bangalore, India | Software Developer – Data Engineering**June 2022 – July 2023**

- Developed core PySpark applications to process massive transactional datasets (high-volume sales and clickstream events), supporting critical analytics and recommendation engines.
- Refactored resource-heavy SQL queries and Spark cluster configurations, boosting data processing throughput by ~30% during peak "Big Billion Day" sale events.
- Deployed partitioning strategies and incremental load logic for daily workflows, which significantly reduced compute overhead and ensured strict SLA adherence.
- Instrumented pipelines with comprehensive logging, job metrics, and alerts, enabling the team to pinpoint failures instantly and reduce mean-time-to-resolution (MTTR).
- Aligned data definitions by working with product and analytics teams to codify key metrics, ensuring consistent reporting across all business dashboards.
- Standardized CI/CD workflows for Spark and SQL code using Git and automated testing, making releases predictable and eliminating manual deployment errors.
- Orchestrated data movement between legacy systems and object storage (S3/Data Lake), tuning resource allocation to handle traffic spikes without system degradation.

BirlaSoft, Hyderabad, India | Software Developer**Nov 2021 – May 2022**

- **Scripted robust data extraction** and transformation logic using **Python** and **SQL** to process structured datasets for internal reporting, ensuring high data consistency for stakeholders.
- **Automated quality assurance workflows** by constructing custom validation rules that identified insufficient data upstream, reducing manual QA effort by ~60%.
- **Reinforced pipeline stability** by integrating structured logging and error-handling into daily jobs, making failure detection and root-cause analysis significantly faster.
- **Resolved production data incidents** by investigating server logs and reproducing errors in **Linux** environments, minimizing downtime for critical processing jobs.
- **Authored technical runbooks** and data flow documentation to assist analysts and junior engineers, standardizing knowledge transfer for everyday operational tasks.
- **Refined data models** and transformation logic to simplify reporting tables, improving reliability for downstream business users.

EDUCATION**Master of Science in Data Analytics**

Indiana Wesleyan University, Marion, Indiana

Bachelor of Technology in Computer Science and Engineering

Kakatiya Institute of Technology and Science, Warangal, India

May 2022